Installation Manual
Version 1.0
Solomon Islands' Integrated Monitoring, Reporting and Verification (iMRV) Tool

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Value based, innovative & sustainable solutions

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About the Manual

Welcome to the Installation Manual Guidebook. This manual is designed to provide clear, step-by-step instructions to help you successfully install and set up **Solomon Islands' Integrated Monitoring, Reporting and Verification (iMRV) Tool** on **FRAPPE FRAMEWORK**. Whether you're a professional installer or a first-time user, this manual covers all the essential information you need to ensure a smooth and efficient installation process.

Inside, you will find system requirements, directory structures, installation process step-by-step, and troubleshooting tips to assist you at every stage. Our goal is to make your installation experience as straightforward and hassle-free as possible, so you can get up and running quickly and confidently.

This manual has been carefully crafted by experts to ensure accuracy and ease of use. We recommend reading through the entire guide before beginning your installation, so you are fully prepared for each step. Should you encounter any issues, our support team is available to assist you.

For any further information or clarification, please contact to:

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1. Introduction

The **Solomon Islands' Integrated Monitoring, Reporting and Verification (iMRV) Tool** has been developed on the Frappe Framework[1] which is a full stack, batteries-included, web framework written in Python. It comes with a simple yet powerful Object Relational Mapper (ORM) as an abstraction over Create, Read, Update, and Delete (CRUD) operations. The default database is MariaDB[2]. Postgres support is in beta.

1.1 Services

Frappe Framework uses the following services for running the SI-iMRV Tool application:

FRAPPE FRAMEWORK	1. MariaDB (database)
	2. Frappe Framework (Python) (web framework)
	3. Redis (cache + queue)
	4. NodeJS (socketio)
	5. File Server

Frappe framework is based on Python, so it uses the virtualenv to setup isolated environments for multiple Python versions. You can also use it to deploy sites with different Frappe versions.

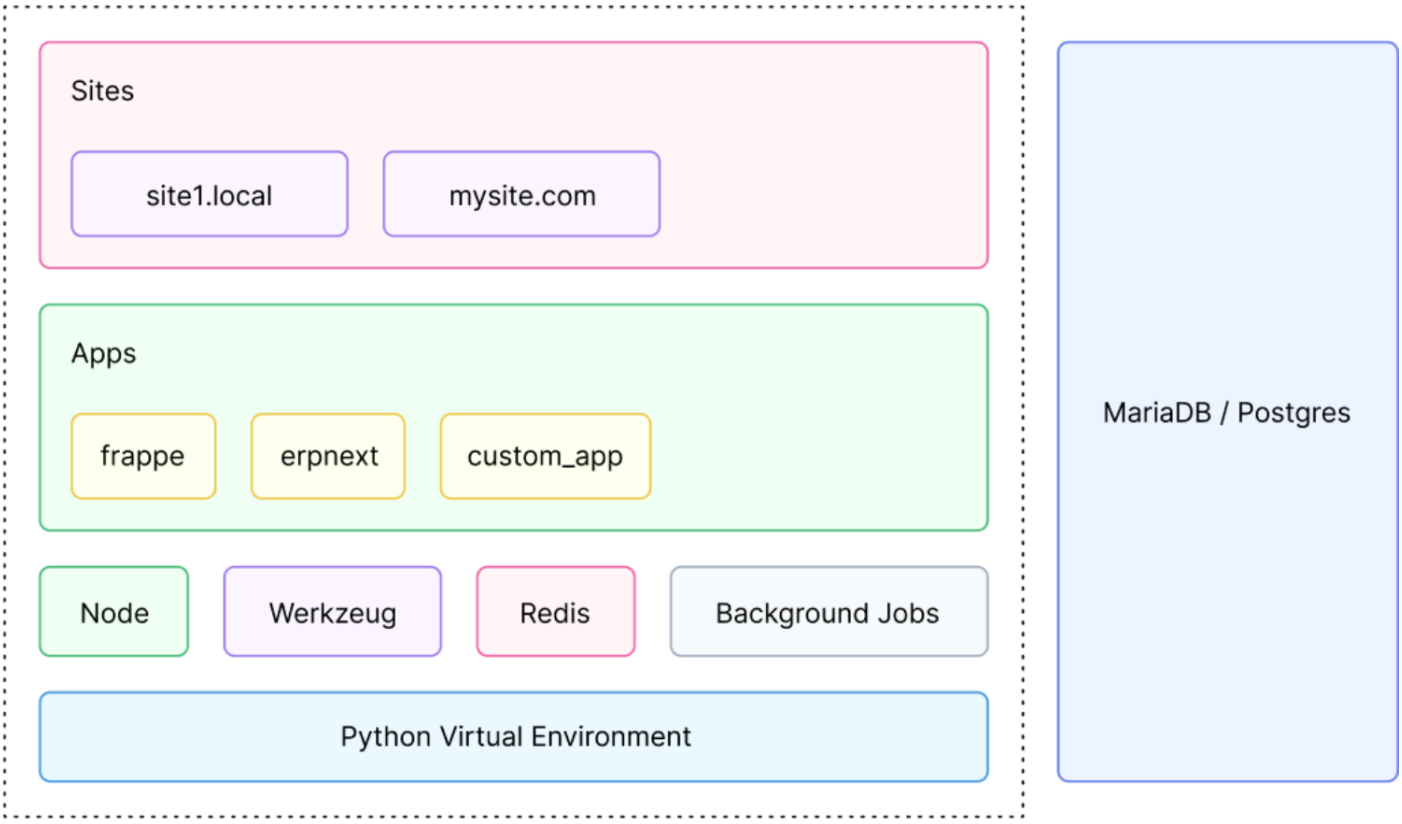
1.2 Frappe Bench

The deployment is handled by Bench, a command line utility that helps install, manage, and run all the services required by Frappe Framework. It also manages site migrations. Bench is also a multi-tenant system where each tenant has its own database and files that are served from separate folders called sites.

[1] https://frappeframework.com/
[2] https://mariadb.org/

Architecture

The following diagram closely resembles the frappe-bench directory structure and its interface with different parts of the stack.



2. System Requirements

2.1. Operating System

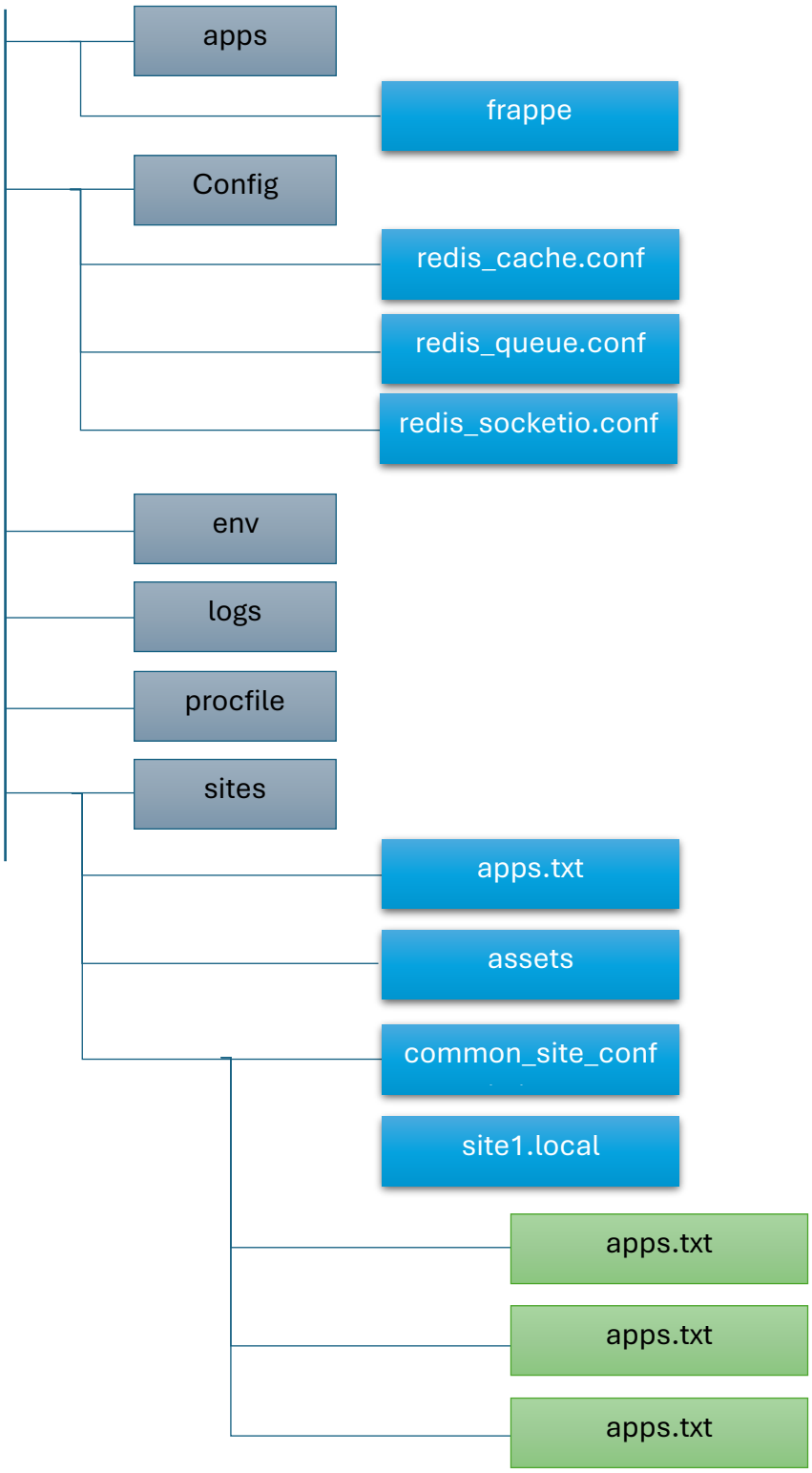
I.	MacOS
II.	Debian / Ubuntu

2.2. Pre-requisites

I.	Oython 2.7 (Python 3.5+ also supported)
II.	MariaDB 10+
III.	Nginx (for production)
IV.	Git
V.	Nodejs
VI.	Yarn
VII.	Redis
VIII.	cron (crontab is required)
IX.	wkhtmltopdf with patched Qt (version 0.12.3) (for pdf generation)

3.Directory structure

When you initialize a new frappe-bench directory, you will have a directory structure similar to:



3.1. Apps

The frappe app and other frappe based apps live in this directory. When you run the command bench new-app app_name, the app will be bootstrapped in this directory. Your custom apps live here, and you are supposed to edit/work with them here.

3.2. Sites

Sites are served from this directory. When you run the command bench new-site site_name, the site will be created in this directory. Sites are distinguished based on their directory name.

3.3. Config

Frappe uses 3 Redis instances to manage caching, job queueing and socketio communication. All those configurations live here.

3.4. Env

The Python virtual environment lives in this directory. Frappe based apps and Python package dependencies are installed here.

For more information, please refer following link:

https://frappeframework.com/docs/user/en/basics/directory-structure

4.App directory structure

We can create a custom app in frappe framework and to create our custom app run the “**bench new-app custom_app**” under the bench folder.

You will be prompted with details of your app, fill them up and an app named “**custom_app**” will be created in the apps folder.

Your app directory structure should look something like this:

apps/custom_app

```
├── README.md
├── custom_app
│   ├── hooks.py
│   ├── custom_app
│   │   └── __init__.py
│   ├── modules.txt
│   ├── patches.txt
│   ├── public
│   │   ├── css
│   │   └── js
│   ├── templates
│   │   ├── __init__.py
│   │   ├── includes
│   │   └── pages
│   │       └── __init__.py
│   └── www
└── pyproject.toml
```

Where,

public: Store static files that will be served from Nginx in production

templates: Jinja templates used to render web views

www: Web pages that are served based on their directory path

custom_app: Default Module bootstrapped with app

modules.txt: List of modules defined in the app

patches.txt: Patch entries for database migrations

hooks.py: Hooks used to extend or intercept standard functionality provided by the framework

pyproject.toml: Specifies how your app is built, you can optionally add 3rd party Python dependencies here which will get installed when your app is installed.

5.Site Directory Structure

Database driven multi-tenant architecture easily lets you host multiple sites on a single server. Each site comes with a database. They may be customized through site specific scripting or Apps may be installed on them.

To create a new site, run the following command from the frappe-bench directory: **bench new-site custom_site**

This command will create a new database, so you need to enter your MySQL root password. It will also ask to set the password for the Administrator user, just set a password that you won't forget.

The site directory structure will look something like this: **sites/custom_site**

```
├── indexes
│   └── web_routes
├── locks
├── logs
├── private
│   ├── backups
│   └── files
├── public
│   └── files
└── site_config.json
```

The indexes folder has indexes generated via Website Search.

The locks folder maintains file-based locks over in-site documents as well as indicators of the state of the site itself.

As you can see, the private folder will contain any database backups and private files. Private files are user uploaded files that need authentication to be accessible.

The public folder will contain files that are accessible without authentication. This can contain website images that should be accessible without login.

The site_config.json file contains configuration that is specific to this site which should not be version controlled. This is like an environment variable file. If you look at the contents of the file, it contains the database configuration values for this site.

To install our Custom App on our site, run the following command

bench --site custom_site install-app custom_app

To confirm if the app was installed, run the following command

bench --site custom_site list-apps

You should see a list of installed apps on your site.

6.Installation

For set up new system please find the following steps

STEP I Install the frappe framework by using the following link https://frappeframework.com/docs/user/en/installation

STEP II After installing frappe, go to the bench folder and create new site with following command

bench new-site site_name

This command will create a new database, so you need to enter your MySQL root password. It will also ask to set the password for the Administrator user, just set a password that you won't forget

STEP III Need to install the custom apps which are mentioned in the git urls document by using following command

A. frappe_side_menu : https://github.com/rajeshscs/MRV-Side-Menu-Solomon-Islands.git
bench get-app https://github.com/rajeshscs/MRV-Side-Menu-Solomon-Islands.git

B. MRV Tool App: https://github.com/rajeshscs/MRV-Solomon-Islands.git
bench get-app https://github.com/rajeshscs/MRV-Solomon-Islands.git

STEP IV Then install the app with following command:

A. bench –site site_name install-app frappe_side_menu

B. bench –site site_name install-app mrvtools

STEP V After installed the apps, you can run the bench by bench start command.It will start the application in specific port, please find the following screenshot for reference

STEP VI To enable the production mode

- **bench config dns_multitenant on**
- **sudo bench setup production** (system_user_name)

- **bench setup nginx**
- **sudo service nginx restart**

STEP VII After the enabling the production mode, to add domain run the following

bench setup add-domain "domain_name(eg:admin.farmlink.in)"

It will ask the site name for which site domain has map

STEP VIII To install SSL using Lets Encrypt

sudo snap install core

If error with sudo: snap: command not found then **sudo apt-get install snapd**

sudo snap refresh core

sudo snap install --classic certbot

sudo ln -s /snap/bin/certbot /usr/bin/certbot

After the installing the certbot need to install the ssl for specific domain

sudo certbot –nginx

Specify the domain to which SSL is to be installed. After need to restart the nginx service

sudo service nginx restart